

R.C. MYTHREYA RAGHAV

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EDUCATION

Bachelor of Engineering, Electrical Engineering

McGill University, Montreal, QC

Sept 2022 – May 2026

Diploma of Collegial Studies (Pure and Applied Science)

Marianopolis College, Montreal, QC

Sept 2020 - Jun 2022

TECHNICAL SKILLS

Programming and Engineering Tools: Java, C, ARM assembly, VHDL, Python, Altium, KiCad

Numerical Software: LabView, MATLAB

Simulation Tools: SIMULINK, LTSpice, Logisim-Evolution

ENGINEERING EXPERIENCE

Digital Design and VHDL Development

- Developed and verified digital designs on Altera FPGAs using Quartus Prime, ensuring robust hardware timing closure.
- Collaborated with team members to simulate, implement, and verify design functionality through model-based testing.

Microelectronics

- Understanding of single-stage, differential, and multistage IC amplifiers, including biasing techniques.
- Designed a multi-stage amplifier circuit consisting of transistors and passive elements to meet a set of specifications.

Digital Sampling and Testing of Mixed-Signal Systems

- Understanding of design methodologies for DIB-based test interfacing between DUTs and ATE, with consideration of circuit parasitics.
- Coverage of DSP concepts relevant to testing ADC/DAC systems, including sampling theory and frequency domain analysis.

Embedded and Digital Systems

- Designed and implemented embedded software interfacing with peripheral devices using I2C communication.
- Gained experience in circuit modeling, timing analysis, asynchronous circuits, and design for testability.

PROJECTS

16-bit CPU Design (Logisim Evolution)

- Designed and implemented a 16-bit CPU using Logisim Evolution, focusing on instruction set architecture (ISA), control unit design, and memory management.
- Developed and tested key components such as ALU, registers, and clocked logic to ensure proper data flow and operation of the CPU.

ESP-32 Based Embedded Microcontroller ([GitHub Link](#))

- Designed an ESP-32 based microcontroller board featuring Wi-Fi and Bluetooth connectivity for wireless embedded applications.
- Implemented hardware integration with onboard IMU and I2C expansion capability for external sensor interfacing.

Gesture Based Musical Instrument (STM32 Microcontroller)

- Developed a gesture-controlled musical instrument in C using an STM32 board and a buzzer, mapping board movements to five distinct musical notes generated using DAC.
- Implemented note recording and playback functionality using memory storage and user input via push buttons.

Wind Estimation System for VTOL Drones (McGill's Drone Team)

- Designed a real-time wind estimation system using ultrasonic transducers and PLLs to improve hovering stability in VTOL drones.
- Proposed a PCB-Based hardware solution for interfacing sensors with a microcontroller, enabling in-flight wind data acquisition.